A PIPELINE FOR INTERPRETABLE NEURAL LATENT DISCOVERY

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ABSTRACT

Mechanistic understanding of the brain requires interpretability of large-scale neuronal computations. Many latent variable model approaches applied to such datasets excel at decoding but yield opaque latent spaces. We address this with NLDisco, a pipeline for interpretable neural latent discovery. Motivated by successful applications of sparse dictionary learning in AI mechanistic interpretability, NLDisco encourages hidden layer neurons in sparse encoder-decoder models to learn interpretable structure in dense neural activity. A flexible, open-source, user-friendly software package supports interpretable latent discovery across recording modalities and experimental paradigms. We validate the pipeline on both synthetic and real extracellular recordings across species and brain regions, showing that it recovers ground-truth features in synthetic data and reveals meaningful representations in real data. These results highlight the pipeline's potential to advance neuroscientific discovery and provide a foundation for future methodological extensions and applications in interpretable modeling of neural activity.

1 INTRODUCTION

Advanced extracellular recording technologies now enable neuroscientists to capture unprecedented volumes of high-dimensional neural[1] data at single-cell resolution (Steinmetz et al., 2021; Raducanu et al., 2017; Cai et al., 2016; Villette et al., 2019; Ouzounov et al., 2017; Ahrens et al., 2013). Understanding the computational principles driving neural activity requires extracting interpretable features from this data. Traditional dimensionality reduction techniques, while useful for visualization and basic analyses, often make invalid assumptions about or altogether fail to disentangle distributed representations, and as a result typically cannot provide a mechanistic understanding of the underlying computations (Cunningham & Yu, 2014; Humphries, 2021). Recent work has produced promising latent variable model (LVM) approaches capable of identifying low-dimensional subspaces that can accurately decode aspects of behavior and environment (Song et al., 2025; Schneider et al., 2023; Le & Shlizerman, 2022; Keshtkaran & Pandarinath, 2022; Yu et al., 2009; Macke et al., 2011; Gao et al., 2016; Low et al., 2018; Jensen et al., 2020; Hernandez et al., 2020; Kim et al., 2021; Hurwitz et al., 2021; Schimel et al., 2022; Kudryashova et al., 2023; Ye et al., 2023; Gondur et al., 2024; Pellegrino et al., 2024; Sani et al., 2024; Pals et al., 2024; Zhang et al., 2024; George et al., 2025; Perkins et al., 2025; Schmutz et al., 2025). However, these methods typically value decoding accuracy over latent interpretability, and consequently all have at least one of the following limitations: poorly interpretable latents, lossy mapping to and/or poor reconstruction of neural data from latent space, priors on the latent and/or neural data space, requirements of additional behavioral, environmental, and/or trial-structured neural data, supralinear scaling with respect to dataset size, or high implementational complexity.

We address each of these limitations in NLDisco, which provides a simple, user-friendly library for interpretable latent discovery from high-dimensional neural data. We consider a latent's interpretability in two key aspects: 1) its correspondence to a specific external variable – a "natural" behavioral or environmental feature[2]; 2) its explicit composition from contributing neural activity. Unlike other

[1]By default, "neural" and "neuron" will refer to neurobiology; e.g. we call biological neuronal spike data "neural activity". In contrast, activity and neurons from artificial neural networks will be explicitly prefaced; e.g. "artificial neural activity" or "the model's neurons".

[2]A "feature" is a sufficiently interpretable latent. We illustrate this in Results.

approaches that require a search for meaningful directions or dynamics in a latent space, NLDisco outputs individual latents that can be directly assessed for interpretability. Additionally, while we showcase NLDisco on spike data, it can be readily used with virtually any other neural recording modality as the only data requirement is any predefined spatiotemporal binning.

NLDisco trains shallow, overcomplete, sparse encoder-decoder (SED) neural network models (Olshausen & Field, 1997; Vincent et al., 2008) whose individual dictionary elements – hidden layer neurons – ideally represent interpretable latents. This particular LVM approach has had many successes in the field of AI mechanistic interpretability, in which such models have typically been implemented as sparse autoencoders, transcoders, or crosscoders (Lindsey et al., 2024; Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024; Dunefsky et al., 2024; Ameisen et al., 2025; Lindsey et al., 2025). For neural data, SEDs are additionally motivated by the principle that we need not explicitly impose structure on the latent space nor model its dynamics to discover interpretable variables.

Specifically, NLDisco bypasses the need to enforce non-Markovian dynamics, which are often a property of an incomplete observation of a system rather than the system itself; any dynamical system can be represented as Markovian via sufficient state augmentation (Takens, 1981). Given this, even methods that appear non-Markovian can be seen as attempts to approximate a complete underlying state. For example, the forward dynamics of a recurrent neural network (RNN) are Markovian in its hidden state (Sussillo & Barak, 2013; Goodfellow et al., 2016), and generalized linear models (GLMs) with history filters (Pillow et al., 2008; Truccolo et al., 2005) use past activity to augment the present state. This perspective aligns with several findings in neuroscience, from activity in motor cortex reflecting a low-dimensional, implicitly Markovian dynamical system (Churchland et al., 2012b; Shenoy et al., 2013), to predictive-coding formulations that posit that the brain maintains an internal state sufficient to predict sensory streams, i.e., a Markovian latent process (Rao & Ballard, 1999; Doya et al., 2007; Friston, 2010). Grounded in this perspective, NLDisco forgoes the challenge of modeling intricate dynamics and instead directly extracts meaningful constituent features of the underlying neural state (Figure S1).

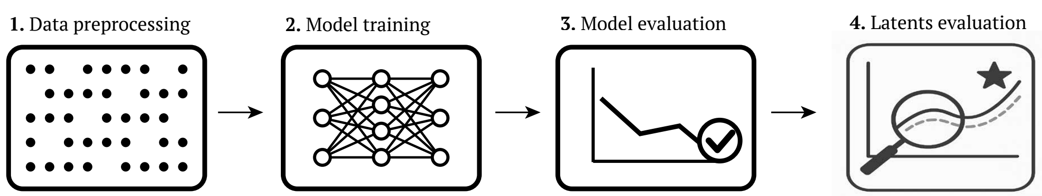

Figure 1: The NLDisco pipeline.
The NLDisco pipeline has 4 stages: 1) Spatiotemporal binning and processing of neural data; 2) Training a model; 3) Evaluating the model; 4) Evaluating the model's latents for feature interpretability. Steps 1-3 can be either semi- or fully-automated.

2 Methods: The NLDisco pipeline

The NLDisco pipeline transforms high-dimensional neural data into a set of interpretable latents in four stages (Figure 1), the first three of which can be fully automated.

2.1 Data preprocessing

The pipeline's first stage prepares neural data for model training. NLDisco includes utilities to process outputs from common spikesorters (e.g. Kilosort (Pachitariu et al., 2016)) by aggregating, binning, and normalizing spike times into a 2D matrix of time by space, in which each bin contains neural activity from a neural unit for a specific time period. Normalization operations include min-max and z-scoring, applied across either the spatial or temporal dimension. This processing is readily adaptable to other forms of acquired neural data, such as ROIs from calcium imaging data processed by tools like Suite2p (Pachitariu et al., 2017). Alternatively, users may pass their own preprocessed neural data directly to the model training stage.

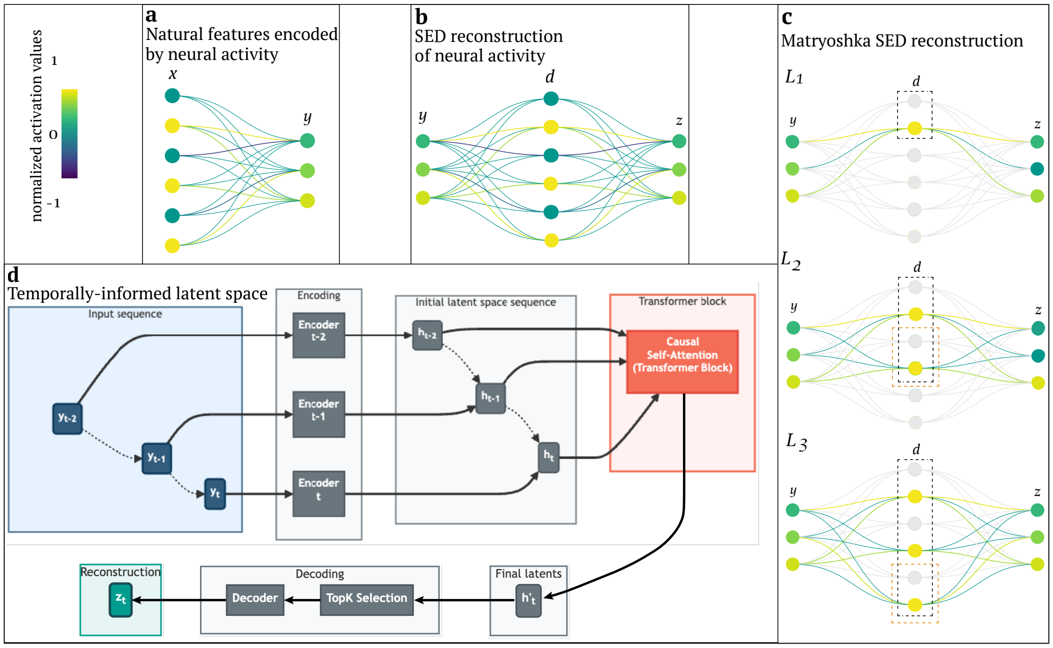

Figure 2: Model architecture considerations
(**a**) Natural, "real-world" features x are encoded by neural activity y. In this example, three active features are simultaneously represented by the joint activity of three neurons. (**b**) A SED reconstructs neural activity z based on y via sparse dictionary elements d. When training is successful, d corresponds to x: sparse dictionary elements (i.e. model neurons) represent natural features. If z tries to recreate y exactly ($\hat{y}$), the model is an autoencoder; in other scenarios (e.g. z is separate but dependent on or related to y) it is a transcoder. (**c**) A Matryoshka SED segments the latent space into multiple nested levels, each of which attempts to do a full reconstruction of the target neural activity. The black boxes indicate the latents involved in a single level, while the light-red boxes indicate the additional latents used at lower-levels. In this example, k=1 for top-k selection of latents to recruit for reconstruction at each level (the yellow neuron within each light-red box). Latents in the highest-level (L_1) will often correspond to high-level features (e.g. a round object), while latents exclusive to the lowest-level (L_3) will often correspond to low-level features (e.g. a basketball). (**d**) Incorporation of a transformer block with sequence input allows imbuing the latents with temporal information corresponding to the evolution of the sequence, which can lead to improved reconstruction and interpretability. Each sample in the input sequence is transformed by the same encoding dictionary matrix in parallel to yield a latent space sequence. Causal self-attention is then performed on this sequence of latents via a single, small multi-head transformer block, yielding a final latent space. The latents are sparsified via top-k and transformed by the decoding dictionary matrix to yield the neural data reconstruction, as in the single, non-sequential input case.

2.2 Model Training

In stage 2, a SED model is trained to reconstruct the processed neural data from latent, sparsely active dictionary elements – the model's hidden layer neurons[3]. The end goal is for these latents to represent disentangled, interpretable representations[4] encoded by the neural activity (Figure 2a,b). Sparsity encourages the model to discover a monosemantic dictionary, in which each latent corresponds to a single feature. For instance, a latent from a model trained on visual cortex activity may correspond to a particular property of a visual stimulus, while a latent from a model trained on motor cortex activity may correspond to a specific aspect of movement.

The model can be configured as an autoencoder (Cunningham et al., 2023) to reconstruct its own input (e.g. V1 activity based on V1 activity), or as a transcoder (Dunefsky et al., 2024) to reconstruct a dependent or related target of its input (e.g. V2 activity based on V1 activity, or V1 activity on day 2 based on V1 activity on day 1). These configurations allow for examining the intrinsic representational structure of a neural population and how representations may be transformed across brain regions and differ over time or between subjects. In each case, the model is trained to minimize the difference

[3]In a SED model: model neuron = dictionary element = latent
[4]In our vernacular: interpretable latent ≈ feature ≈ representation

between the reconstructed and target neural data. The full training procedure is summarized in Model training procedure. Here we discuss three key components: the batch top-k sparsity, Matryoshka architecture, and optional integration over latent space history via self-attention.

A batch top-k operator preserves only the $k \cdot B$ latents with the highest activation magnitudes across a batch of size B (Bussmann et al., 2024). In contrast to indirect approaches (e.g. an L1 penalty), batch top-k directly controls sparsity while flexibly permitting the number of active latents to vary per sample in the batch, aligning with the variable complexity of neural states. A novel variant of the Matryoshka SED architecture (Bussmann et al., 2025) segments the latent space into nested hierarchical levels, where each level attempts a full reconstruction of the target neural activity (Figure 2c). This allows for multi-scale feature representation at single timepoints, and mitigates "feature absorption," a common issue where a more specific feature subsumes a portion of a more general feature (Chanin et al., 2025). For example, a model can learn that a "drifting grating" is a subset of the more general "grating" visual stimulus class without sacrificing the representation of either (Mouse spike data in a passive visual task). Lastly, if input data is provided as a sequence of timebins rather than a single timebin, an optional transformer block can integrate information over the latent space history via self-attention (Figure 2d). This optional mechanism can be viewed as a form of explicit state augmentation; while the underlying Markovian state can theoretically be inferred from a single timebin, autoregression on a latent space sequence can serve as a powerful inductive bias, making it easier for the model to discover features that unfold over longer timescales.

Effectively searching the large and flexible model space created by these architectural options is crucial, as we empirically find that these components can significantly improve both reconstruction accuracy and feature interpretability (Model architecture). To streamline this search with minimal manual specification, NLDisco supports fully automated training via hyperparameter sweep configurations that support local or distributed training across one or multiple CPUs or GPUs (Hyperparameter sweeps).

2.3 Model evaluation

Following training, model quality is evaluated using a suite of quantitative metrics. NLDisco adapts domain-general core metrics from the established mechanistic interpretability benchmark SAEBench (Karvonen et al., 2025), and introduces others tailored for neural data. This stage serves as a critical checkpoint: while these metrics do not directly assess latent interpretability, they allow a researcher to determine if the model has learned a sparse, accurate, robust representation of the neural data. A model that performs well across these metrics is a strong candidate for the subsequent, final pipeline stage of evaluating individual latents.

Model evaluation centers on the trade-off between reconstruction fidelity and latent dictionary sparsity. High reconstruction fidelity is trivial to achieve with a large and dense code but undermines the goal of finding disentangled, interpretable latents. NLDisco therefore first reviews dictionary quality via latent sparsity metrics (Figure S2 top). The mean L0 norm measures the average number of active latents per sample, set indirectly by the batch top-k parameter, while the latent activity density visualizes the firing distribution across all latents. Together these metrics reveal whether the model has learned an efficient code that uses many latents intermittently while avoiding common pitfalls such as "representational collapse" into too many constantly active latents and/or widespread "feature death", in which a large portion of the dictionary never activates. These sparsity metrics are complemented by a couple of reconstruction fidelity metrics: variance explained (R^2) and cosine similarity of reconstruction-to-target neural activity across both spatial and temporal dimensions (Figure S2 bottom). High scores in these metrics confirm that the sparse code is a sufficient representation that has captured salient information present in the target neural data.

For a deeper diagnosis, several optional metrics are available. To check whether explanatory power is well-distributed, a variance attribution analysis quantifies the overall reconstruction variance explained by each individual latent. Additionally, to check if temporal information present in the target neural data is preserved by the latent reconstruction, a spectral frequency analysis compares the reconstruction's frequency content to that of the target neural data, given a specified duration.

Overall, these metrics provide a high-level overview of model quality, and to facilitate bespoke evaluation, the model's raw parameter, gradient, and activation data during training and evaluation are accessible.

2.4 LATENTS EVALUATION

Lastly in stage 4, the latents produced by the model are evaluated for interpretability. This can include visualizing their activation patterns over time and experimental conditions, assessing their decoding performance, and tracing each latent back to the biological units that contribute to it.

In the analyses reported here, latents are mapped to behavioral and environmental variables using a selectivity score that compares the latent's activation probability within and outside a specified condition. A score of 0.5 indicates no preference, while larger values indicate greater condition-specific selectivity. The full definition is given in Selectivity score, and the resulting latent-evaluation approaches are demonstrated in Results.

3 RESULTS

We evaluate NLDisco on four open-source datasets that span multiple species, brain regions, and experimental paradigms:

1. Simulated single-unit spike data of rat hippocampus during a navigation task (Simulated rat hippocampal spike data in a navigation task).
2. Single-unit spike data from macaque motor cortex during an active reaching task (Macaque motor cortex spike data in an active reaching task).
3. Single-unit spike data from mouse retrosplenial cortex and superior colliculus during an ethological foraging assay (Mouse spike data in an ethological foraging assay).
4. Single-unit spike data from mouse visual cortex during a passive visual task (Mouse spike data in a passive visual task).

To benchmark NLDisco, we compare it against several other prominent neural LVM approaches. For fair comparisons focused on the main goal of unsupervised interpretable latent discovery, we detail paper-published methods (Comparison of NLDisco with other neural LVM methods) that:

1. Prioritize an interpretable latent space (as opposed to, for instance, solely strong decoding performance).
2. Don't require multimodal nor trial-structured data.
3. Have published results on the Churchland et al. (2012a) dataset.

Among these, we visualize results from LangevinFlow (Song et al., 2025) and CEBRA (Schneider et al., 2023), as they represent the current top-performing autoencoder and non-autoencoder methods, respectively, on Neural Latents Benchmark (Pei et al., 2022). We also include (sparse)NMF and PCA as widely-used dimensionality reduction baselines.

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- On the synthetic dataset, we show that NLDisco can recover all ground-truth latents, and use them to decode natural features of interest.

- On one natural dataset, we compare NLDisco to other popular approaches that have also been applied to the same dataset, and show that NLDisco more clearly reveals certain features than these other methods, while maintaining high reconstruction and decoding accuracy.

- On the other two natural datasets, we show that NLDisco can reveal expected and unexpected features (optinally / time-permitting, that CEBRA and LangevinFlow don't find), highlighting its use as a powerful tool for neuroscientific discovery.

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3.1 SIMULATED RAT HIPPOCAMPAL SPIKE DATA IN A NAVIGATION TASK

Following Theodoni et al. (2018), we simulated the activity of hippocampal neurons during a navigation task in a virtual linear environment (see Simulated rat spike data in a navigation task for full simulation details). The virtual rat moved along a 1-m linear track with stochastic velocity.

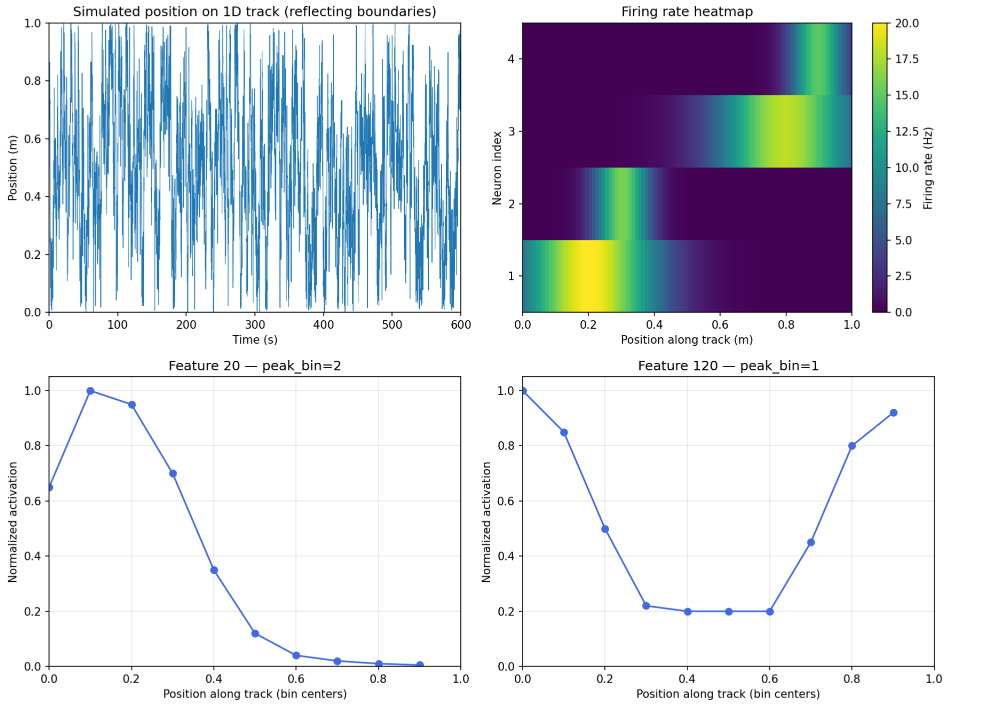

Figure 3: Features identified from simulated place-cell activity.
Simulated place-cell activity and learned latent representations on a 1-m linear track. Top-left: the simulated positional trajectory. Top-right: the firing-rate tuning curves for four place cells. Bottom-left: a single-place-cell-like feature active at an early position on the track. Bottom-right: a multi-place-cell-like feature active near the beginning and end of the track.

We modeled four place cells with heterogeneous fields and, to determine whether NLDisco could discover task-relevant features within larger neural population activity, added activity from 96 "noise" neurons independent of position.

We trained an SED with 128 latents and a sparsity constraint of four active latents per time bin. While some latents captured random noise fluctuations, others successfully learned spatial signals. Two classes of spatial representations emerged: single-place-cell-like features and multi-place-cell-like features (Figure 3), demonstrating NLDisco's ability to isolate distinct functional representations.

3.2 Macaque motor cortex spike data in an active reaching task

To test NLDisco on real neural data, we used the `churchland_shenoy_neural_2012` dataset (Churchland et al., 2012a) prepared by Brainsets (Brainsets Project, 2022), also referred to as MC_Maze (Neural Latents Benchmark Team, 2022). In this dataset, monkeys perform a reaching task under a variety of maze configurations. Recordings contain roughly 200 neurons per monkey, spanning dorsal premotor (PMd) and primary motor (M1) cortices, alongside behavioral measures including hand position, velocity, acceleration, and gaze position.

To identify features, SED latents were automatically mapped to metadata variables of interest using the selectivity score defined in Selectivity score. This approach revealed a wide range of features, including latents tuned to different ranges of hand velocity (Figure 4a). The same feature types emerged independently across subjects and generalized to unseen sessions.

We also identified features aligned with environmental variables, such as the hit target position (Figure 4b). Beyond mapping latents to behavioral or task variables, NLDisco allows features to be traced back to their underlying source units (Figure 4c), providing both correspondence and compositional interpretability.

Figure 4: Features identified in the Churchland dataset.
(a) Examples of SED features tuned to a continuous behavioral variable: hand velocity. The dashed line marks 0.5, corresponding to equal activation probability inside and outside a bin (no selectivity), while values above 0.5 indicate condition-specific tuning. Features shown are selective for low, high, and "extreme" velocities (active at both ends of the range). **(b)** Example of an SED feature tuned to a discrete environmental variable: the hit target being on the right side of the workspace (positive x-coordinate). Selectivity scores are computed as in (a), with the dashed line again marking the non-selective baseline at 0.5. **(c)** Mean z-scores of biological neurons when the right-side target feature (latent 169) is active, showing which units in the recorded population are systematically co-active with the latent.

3.3 Mouse spike data in an ethological foraging assay

- We analyze data from the Aeon project, which provides continuous long-term recordings of freely behaving mice in enriched environments. This dataset allows us to study neural dynamics during naturalistic behaviors over extended time periods, from minutes to weeks.

- Brief dataset info...

- Features found and decoding accuracy comparisons with LangevinFlow and CEBRA...

Overall, these results demonstrate that popular neural LVM approaches often fail to find interpretable latents, while NLDisco can robustly discover discrete and continuous, environmental and behavioral feature representations across varying timescales and levels of abstraction. Importantly, same feature representations can be found across subjects and are stable within a subject across time. This highlights the interpretable latent as a fundamental unit of representation, enabling a direct mapping from representation to source neuronal activity.

All results are reproduced in publicly available notebooks – see Software and data availability.

4 Discussion

4.1 Contributions

This paper makes the following key contributions:

1. We introduce a pipeline for applying sparse dictionary learning methods specifically designed to discover interpretable neurobiological latents ("features").
2. Our pipeline is end-to-end from spatiotemporally binned neural data to interpretable latent discovery: containns functionality for neural data preprocessing, model training, model evaluation, and feature evaluation.

3. We provide systematic evaluation on multiple synthetic and natural large-scale neurobiological datasets, demonstrating the effectiveness of our approach across different species, brain regions, recording techniques, experimental conditions, and spatiotemporal scales.
4. We show that MSAE-derived features enable effective decoding of behavioral and stimulus variables, validating their biological relevance.
5. We release our code open-source with interactive tutorials for facilitating broader adoption in the neuroscience community.

4.2 PROS AND CONS

Advantages:

- **Interpretability**: The sparse and multi-scale nature of the learned features facilitates direct interpretation and comparison with existing neuroscience knowledge.
- **Scalability**: The model architecture is designed to scale to large datasets with thousands of neurons and millions of time points.
- **Flexibility**: The framework can be adapted to various data modalities, including spike trains, calcium imaging, and LFP/EEG signals.

Limitations:

- Model performance is sensitive to the choice of hyperparameters, requiring careful tuning and validation.
- While our model reveals correlational structure, it does not inherently infer causal relationships between neural features and behavior.
- The model is a simplified representation of neural circuits and does not capture all aspects of biological complexity.
- Latent interpretability is subject to human subjectivity.

4.3 FUTURE WORK

- Actually apply to other recording modalities
  - LFP & EEG (reconstruct voltage values from C channels over T time)
  - Calcium imaging (reconstruct dF/F values from R RoIs over T time)
- Actually apply to multimodal data
  - e.g. reconstruct spikes from spikes + multimodal behavior (e.g. pose-tracking, gaze-tracking, etc.)
- Use history to improve reconstructions and feature interpretability?
  - While our codebase optionally allows for using history length, in practice we did not get improved reconstructions nor more interpretable latents, but in theory this approach could be improved to do so.
- CLTs & SCCs
  - Cross-region prediction
  - Brain diffing

AI USE STATEMENT

Generative AI tools were used to assist with manuscript organization, formatting, conference-guideline research, and editing for readability. All AI-assisted work was reviewed by the authors. The authors take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

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To maintain blinding, acknowledgements in this submission are omitted, and will be included in the final version.

5 Appendix

5.1 Software and data availability

To maintain blinding, software and data availability information can be found in the anonymized repository at . . . TODO. See the 'notebooks' folder for reproduction of section 3 and Additional results.

5.2 On interpretable latents

Many neural LVM approaches focus on decoding from a low-dimensional latent space. However, the structure of this latent space can be inherently complex and difficult to interpret. NLDisco bypasses this challenge by learning individual latents directly, as illustrated in the toy data shown in Figure S1.

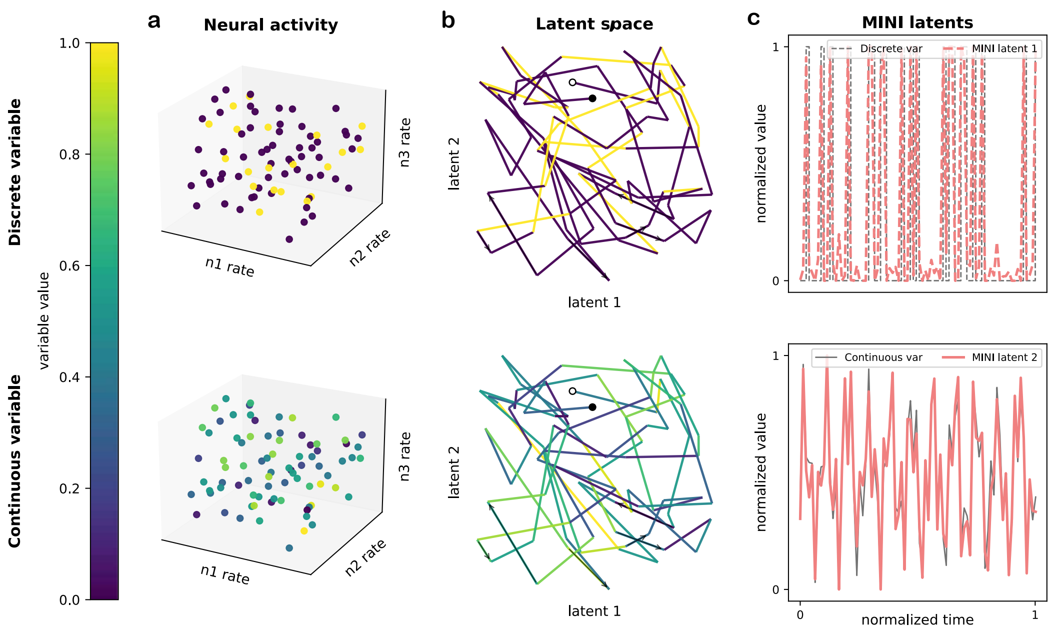

Figure S1: Interpretable latents in a complex latent space.
This toy example highlights the utility of NLDisco. The two rows show two different variables (top: discrete; bottom: continuous), each uniquely encoded by the same underlying neural activity made up of three neurons' firing rates. The colorbar shows the variables' values as a function of this neural activity. (**a**) Each point in the scatterplots represents a moment in time. (**b**) A projection of this activity into a 2D latent space creates tangled trajectories where variable states (e.g. 'on' and 'off' in the discrete case, and 'high' and 'low' in the continuous case) are not easily distinguished. The start and end points of the trajectories are marked by white and black dots, respectively, while arrows indicate trajectory direction. (**c**) In contrast to the tangled latent space, NLDisco finds individual latents corresponding to each variable, demonstrating the potential for improved interpretability of neural representations.

5.3 Selectivity score

Units are mapped to metadata variables through the calculation of a selectivity score. For a latent l and condition c (a variable/value combination, e.g. velocity between 0 and 1):

$$\text{activation\_frac\_during} = \frac{\#\{\text{activations of } l \text{ in examples with } c\}}{\#\{\text{examples with } c\}} \quad (1)$$

$$\text{activation\_frac\_outside} = \frac{\#\{\text{activations of } l \text{ in examples without } c\}}{\#\{\text{examples without } c\}} \quad (2)$$

$$\text{selectivity\_score} = \frac{\text{activation\_frac\_during}}{\text{activation\_frac\_during} + \text{activation\_frac\_outside}} \quad (3)$$

A selectivity score of 0.5 corresponds to no preference, meaning the unit is equally active inside and outside the condition. Scores above 0.5 indicate condition-specific tuning, with larger values reflecting stronger selectivity. Scores below 0.5 indicate the unit is more active outside the condition.

5.4 Additional pipeline details

5.4.1 Model training

Some notes on Model training procedure:

Dead Latent Resurrection. A common failure mode in training SEDs is "latent death," where dictionary elements cease to activate for any input. We address this with an auxiliary loss designed to revive dead latents. We monitor feature activation frequencies and identify a set of dead latents $\mathcal{D}$. These latents are then trained via an auxiliary MSE loss to reconstruct the residual error of the primary model ($\mathbf{x}_{\text{target}} - \hat{\mathbf{x}}_L$). The gradients from this auxiliary loss are exclusively applied to the parameters of the dead latents, thereby encouraging them to learn useful representations without disrupting the training of active latents.

5.4.2 Model architecture

The core of NLDisco is an overcomplete sparse encoder-decoder model with a single hidden layer and ReLU activations. While we experimented with more complex architectures, such as multi-layer decoders, we found that they often made interpreting the resulting latents more difficult without significant reconstruction gains. The Matryoshka architecture is particularly useful for datasets where features are expected to exist at multiple scales, as it helps mitigate "feature absorption," where general features are partially subsumed by more specific ones (Bussmann et al., 2025). The transformer block is most beneficial when analyzing data with meaningful temporal dynamics, as it allows the model to learn features that evolve over a sequence rather than only instantaneous neural patterns.

5.4.3 Model evaluation

Model evaluation centers on the trade-off between reconstruction fidelity and latent dictionary sparsity. High reconstruction fidelity is trivial to achieve with a large and dense code but undermines the goal of finding disentangled, interpretable latents. NLDisco therefore first reviews dictionary quality via latent sparsity metrics (Figure S2 top). The mean L0 norm measures the average number of active latents per sample, set indirectly by the batch top-k parameter, while the latent activity density visualizes the firing distribution across all latents. Together these metrics reveal whether the model has learned an efficient code that uses many latents intermittently while avoiding common pitfalls such as "representational collapse" into too many constantly active latents and widespread "feature death" in which a large portion of the dictionary never activates. These sparsity metrics are complemented by a couple of reconstruction fidelity metrics: variance explained (R^2) and cosine similarity of reconstruction-to-target neural activity across both spatial and temporal dimensions (Figure S2 bottom). High scores in these metrics confirm that the sparse code is a sufficient representation that has captured salient information present in the target neural data.

For a deeper diagnosis, several additional optional metrics are available. To check whether explanatory power is well-distributed, a variance attribution analysis quantifies the overall reconstruction variance explained by each individual latent (TODO). And to check if temporal information present in the

Algorithm 1 Model training procedure

Data definitions:

  Input neural data: $\mathbf{Y} \in \mathbb{R}^{B \times S \times N_{\text{in}}}$ (Batch, Sequence length, Input neural units)

  Target neural data: $\mathbf{Z} \in \mathbb{R}^{B \times N_{\text{out}}}$ (Batch, Output neural units)

Model definitions:

  Encoder: $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{D_{\text{max}} \times S \times N_{\text{in}}}$, $\mathbf{b}_{\text{enc}} \in \mathbb{R}^{D_{\text{max}}}$ (Hidden layer neurons)

  Decoder: $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{N_{\text{out}} \times D_{\text{max}}}$, $\mathbf{b}_{\text{dec}} \in \mathbb{R}^{N_{\text{out}}}$

  Transformer block (optional) $\theta_{\mathbf{t}}$: $\{\mathbf{W}_{Q,K,V} \in \mathbb{R}^{D_{\max} \times D_{\max}}, \ldots\}$

  Matryoshka levels: $\{\mathbf{D_l}\}_{\mathbf{l=1}}^{\mathbf{L}}$

  Weights each level's reconstruction loss (optional): $\{\lambda_\mathbf{l}\}_{\mathbf{l=1}}^{\mathbf{L}}$

  Auxiliary loss weight: γ

procedure TRAIN_STEP($\mathbf{Y}, \mathbf{Z}$)

  ▷ *— Forward Pass —*

  $\mathbf{A} \leftarrow \text{ReLU}(\mathbf{Y}\mathbf{W}_{\text{enc}}^T + \mathbf{b}_{\text{enc}})$ ▷ Get encoder activations

  if $S > 1$ **then**

    $\mathbf{A} \leftarrow \text{SelfAttention}(\mathbf{A}; \theta_{\text{attn}})$ ▷ Temporally integrate over latent space

  end if

  $\mathcal{L}_{\text{recon}} \leftarrow 0$

  for $l = 1$ to L **do** ▷ For each level...

    $\hat{\mathbf{A}}_l \leftarrow S_{\text{topk}}(\mathbf{A}_{:,:D_l})$ ▷ Sparsify latents with batch top-k

    $\hat{\mathbf{Z}}_l \leftarrow \text{ReLU}(\hat{\mathbf{A}}_l \mathbf{W}_{\text{dec},:,:D_l}^T + \mathbf{b}_{\text{dec}})$ ▷ Reconstruct target (decoder activations)

    $\mathcal{L}_{\text{recon}} \leftarrow \mathcal{L}_{\text{recon}} + \lambda_l \cdot \text{MSLE}(\mathbf{Z}, \hat{\mathbf{Z}}_l)$ ▷ Get reconstruction loss

  end for

  ▷ *— Auxiliary Loss for Dead Latent Resurrection —*

  $\mathcal{D} \leftarrow \text{GetDeadLatents}(\mathbf{A})$

  if any($\mathcal{D}$) **then** ▷ Only compute if dead latents exist

    $\mathbf{R} \leftarrow \mathbf{Z} - \hat{\mathbf{Z}}_L$ ▷ Get reconstruction residual

    $\hat{\mathbf{R}} \leftarrow \text{ReLU}((\mathbf{A} \odot \mathcal{D})\mathbf{W}_{\text{dec}}^T + \mathbf{b}_{\text{dec}})$ ▷ Reconstruct residual from dead latents

    $\mathcal{L}_{\text{aux}} \leftarrow \text{MSE}(\mathbf{R}, \hat{\mathbf{R}})$ ▷ Get auxiliary loss

  else

    $\mathcal{L}_{\text{aux}} \leftarrow 0$

  end if

  $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{recon}} + \gamma \mathcal{L}_{\text{aux}}$ ▷ Total loss: reconstruction + auxiliary

  ▷ *— Backward Pass & Parameter Update —*

  $\mathbf{g} \leftarrow \nabla_\theta \mathcal{L}_{\text{total}}$ ▷ Compute gradients, masking $\nabla \mathcal{L}_{\text{aux}}$ to dead latents

  $\theta \leftarrow \text{Update}(\theta, \mathbf{g})$ ▷ Update model parameters

end procedure

target neural data is preserved by the latent reconstruction, a spectral frequency analysis compares the reconstruction's frequency content to that of the target neural data, given a specified duration (TODO).

5.4.4 Latents evaluation

In this paper, we used prepared dashboards to visualize latent activity in the context of the simulated spatial-navigation task and the in vivo motor-control reaching task. More generally, when working with new datasets, we recommend designing similarly simple, bespoke dashboards that place latent activity alongside relevant environmental and behavioral variables.

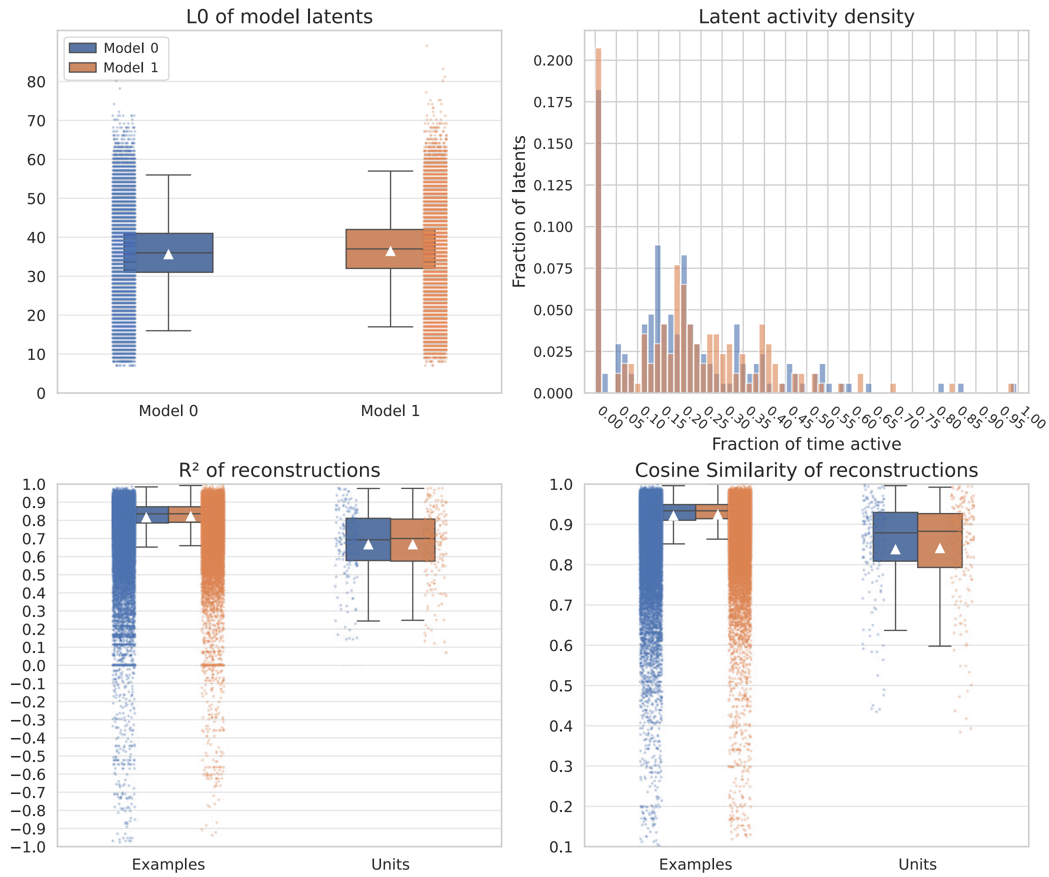

Figure S2: Model evaluation metrics.
This example SAE model evaluation examines a three-level Matryoshka architecture with a total of 320 latents and top-k selections of 12 $\cdot B$, 24 $\cdot B$, and 36 $\cdot B$ active latents per batch of size $B = 1024$. The recording contained 200 units, and the model was trained on 182,868 samples of 50-ms binned spike counts. The model exhibits an approximately normal distribution of the mean latent L0 norm, few dead or tonically active latents, and high neural reconstruction fidelity as measured by R^2 and cosine similarity across both units and examples, all indicators of healthy training.

5.4.5 Hyperparameter sweeps

Effective hyperparameter tuning is crucial for balancing reconstruction fidelity with latent interpretability. We typically sweep over key architectural and optimizer parameters. Architectural choices center on the Matryoshka configuration: `dsed_topk_map` controls the sparsity at each hierarchical level, while `dsed_loss_x_map` sets the relative weight of each level's reconstruction in the total loss. We also tune `dead_latent_window`, which determines when an inactive latent is revived via an auxiliary residual-reconstruction loss, as well as standard optimizer parameters such as learning and decay rates.

Although optimal settings are data-dependent, we use several general heuristics. The total number of latents can initially be set relative to the number of training examples (e.g., $n_{\text{examples}}/10$) and adjusted downward based on reconstruction performance. The top-k value for a level should reflect a reasonable estimate of how many features might be co-active in a given time window. In our experiments, a cosine learning-rate schedule and the MSLE loss function often outperformed static learning rates and MSE loss, respectively, and therefore serve as useful defaults.

5.5 Additional results

5.5.1 Simulated rat spike data in a navigation task

The virtual rat moved along a 1-m linear track, with its velocity sampled from an Ornstein–Uhlenbeck (OU) process:

$$dv_t = \theta(\mu - v_t)\,dt + \sigma\,dW_t, \tag{4}$$

with parameters $\theta = 1.0$ (mean reversion), $\mu = 0.0$ m/s (long-run mean), $\sigma = 0.4$ (volatility), and $v_0 = 0.0$ (initial velocity). The simulation lasted 10 minutes (600 s) at a temporal resolution of 100 Hz.

Place fields were modeled as Gaussian functions of position. Two cells had broad fields ($\sigma = 20$ cm) centered at 20 and 80 cm, while two had narrow fields ($\sigma = 10$ cm) centered at 30 and 90 cm. Peak firing rates were 20, 16, 18, and 15 Hz, respectively, with a 0.1-Hz baseline. The 96 noise neurons had stable mean firing rates sampled uniformly between 1 and 5 Hz and were independent of position. Spikes were generated as Bernoulli draws using the firing-rate-by-time-step probability.

The resulting spike-count matrix was used to train an SED with 128 latents and a sparsity constraint of four active latents per time bin.

5.5.2 Real macaque spike data in a reaching task

Dataset info

- Exp metadata & info
  - Session duration
  - Stimulus conditions
- Neural data info
  - Brain regions
  - Number of units
  - Number of spikes
  - Spike summary stats
- Access instructions

Latent comparisons with sparseNMF, t-SNE, and UMAP

5.5.3 Real mouse spike data in an ethological foraging task

Dataset info

- Exp metadata & info
  - Session duration
  - Stimulus conditions
- Neural data info
  - Brain regions
  - Number of units
  - Number of spikes
  - Spike summary stats
- Access instructions

Latent comparisons with sparseNMF, t-SNE, and UMAP

5.5.4 Mouse spike data in a passive visual task

In addition to the three main results we showcase in the main text (section 3), here we show NLDisco applied to a visual coding Neuropixels dataset (Siegle et al., 2021) from the Allen institute.

Dataset info

- Exp metadata & info
  - Session duration
  - Stimulus conditions
- Neural data info
  - Brain regions
  - Number of units
  - Number of spikes
  - Spike summary stats
- Recreation instructions

As done in section 3, here we also compare a trained NLDisco model to Cebra and LangevinFlow models...

Latent comparisons with sparseNMF, t-SNE, and UMAP **Stimulus decoding accuracy:**

- Natural image classification: 87.3% (vs. 76.2% for raw data, 81.4% for PCA)
- Motion direction decoding: 94.1% (vs. 88.7% for raw data, 90.2% for ICA)
- Orientation discrimination: 91.6% (vs. 85.3% for raw data, 87.9% for standard SAE)

Behavioral state decoding:

- Locomotion state classification: 88.7% accuracy
- Pupil size (arousal) prediction: Pearson r = 0.73
- Visual attention state: 82.4% accuracy

5.6 Methods Comparison

Here we highlight 15 features of neural LVM methods and create a table displaying how NLDisco and other relevant methods compare against the following features:

- *Requires multimodal data*: Whether the method requires multimodal data (e.g. video data, or various forms of behavioral data, in addition to neural data).
- *Requires trial-structured data*: Whether the method requires trial-structured data.
- *Supports multimodal data*: Whether the method can incorporate multimodal data, even if not required.
- *Supports trial-structured data*: Whether the method can use trial-structured data, even if not required.
- *Learns sparse, easily identifiable latents*: Whether the method by default learns sparse, easily identifiable latents.
- *Learns hierarchical latents*: Whether the method by default learns latents that are hierarchically organized (e.g. whether the method can learn one latent that corresponds to a particular behavior, and another, sparser latent that corresponds to a sub-behavior of the first)
- *Learns temporally precise latents*: Whether the method by default learns latents that are time-locked to neural events at the resolution of the desired event, potentially down to single-spike precision.
- *Learns a continuously-valued latent space*: Whether the method by default learns latents that change smoothly as a function of changes in the input neural data.
- *Can use temporal dynamics to update the latent space*: Whether the method can use temporal dynamics (e.g. neural data or latent space history) to update the latent space.

- *Imposes a prior on the latent space*: Whether the method imposes a predfined structure on the latent space (e.g. geometric constraints like orthogonality of latents, or distributional assumptions like a Gaussian latents).
- *Uses nonlinear dynamics*: Whether the method learns latents from nonlinear neural dynamics.
- *Can be used as a generative model*: Whether the method can be used to generate new neural data samples from the learned latent space.
- *Enforces neural data reconstruction*: Whether the method needs to perform neural data reconstruction when learning latents.
- *Has approximate linear time scaling*: Whether the method has linear time scaling with respect to the number of data points in an example and in the dataset.
- *Requires significant hyperparameter tuning*: Whether the method requires significant hyperparameter tuning to learn interpretable latents.

Light-green text indicates that the method has an ideal implementation of the feature, while dark-red text indicates a shortcoming.

Feature	**NLDisco***	LangevinFlow* (Song et al., 2025)	CEBRA* (Schneider et al., 2023)	ST-NDT (Le & Shlizerman, 2022)	AutoLFADS (Keshtkaran & Pandarinath, 2022)	UMAP (McInnes et al., 2018)	t-SNE (van der Maaten & Hinton, 2008)	sparseNMF** (Hoyer, 2004)	ICA (Comon, 1994)	PCA (Hotelling, 1933)
Requires multimodal data	No	No	No	No	No	No	No	No	No	No
Requires trial-structured data	No	No	No	No	No[^]	No	No	No	No	No
Supports multimodal data	No[†]	No	Yes	No	No	No	No	No	No	No
Supports trial-structured data	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Learns sparse, easily identifiable latents	Yes	No	No	No	No	No	No	Yes	No	No
Learns hierarchical latents	Yes	No	No	No	No	No	No	No	No	No
Learns temporally precise latents	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No
Learns a continuously-valued latent space	No[‡]	Yes	Yes	Yes	Yes	Yes	Yes	No[‡]	Yes	Yes
Can use temporal dynamics to update the latent space	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
Imposes a prior on the latent space	No	Yes	No	No	Yes	No	No	No	Yes	Yes
Uses nonlinear dynamics	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
Can be used as a generative model	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
Enforces neural data reconstruction	Yes	Yes	No	Yes	Yes	No	No	Yes	No	No
Has approximate linear time scaling	Yes[#]	No	Yes	No	No	Yes	No	No	Yes	Yes
Requires significant hyperparameter tuning	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No

Table 1: Comparison of NLDisco with other neural LVM methods.

*: Latent comparison visualized in Results
**: Latent comparison visualized in Additional results
†: Not a fundamental limitation – could be added as a feature
‡: Enforced sparsity can cause step-like jumps in the data-to-latents mapping
#: In the standard implementation, without a transformer layer in the decoder
^: Can bin data to create “pseudo”-trials